Icer Erhan
Problem Set 6

Problem 1

Let X and Y be two binary random variables and $Z = X \oplus Y$

a) $H(Z|X) = \sum_{x \in \mathcal{X}} \sum_{z \in \mathcal{Z}} p_{z,x}(z,x) \log p_{z|x}(z|x) = \sum p_{z,x}(z,x) \log \left(\frac{p_{z,x}(z,x)}{p_X(x)} \right)$

$H(Y|X) = \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p_{y,x}(y,x) \log p_{y|x}(y,x) = \sum p_{y,x}(y,x) \log \left(\frac{p_{y,x}(y,x)}{p_X(x)} \right)$

$\alpha, \beta \in \{0,1\}$ $\quad p(z=\alpha, x=\beta) = p(X \oplus Y = \alpha, x = \beta) = p(Y = \alpha \oplus \beta, x = \beta)$

$\rightarrow$ $p(z=0, x=0) = p(y=0, x=0)$ $\quad (\Leftarrow p(z=0) = p(x=0, y=0) + p(x=1, y=1))$
$p(z=0, x=1) = p(y=1, x=1)$
$p(z=1, x=0) = p(y=1, x=0)$ $\quad (\Leftarrow p(z=1) = p(x=0, y=1) + p(x=1, y=0))$
$p(z=1, x=1) = p(y=0, x=1)$

For each $p_{z,y}$ there is a $p_{y,x}$ different associated. (and reciprocally)

$$H(Z|X) = p(z=0, x=0) \log \frac{p(z=0, x=0)}{p(x=0)} + p(z=0, x=1) \log \frac{p(z=0, x=1)}{p(x=1)}$$
$$+ p(z=1, x=0) \log \frac{p(z=1, x=0)}{p(x=0)} + p(z=1, x=1) \log \frac{p(z=1, x=1)}{p(x=1)}$$
$$= p(y=0, x=0) \log \frac{p(y=0, x=0)}{p(x=0)} + p(y=1, x=1) \log \frac{p(y=1, x=1)}{p(x=1)}$$
$$+ p(y=1, x=0) \log \frac{p(y=1, x=0)}{p(x=0)} + p(y=0, x=1) \log \frac{p(y=0, x=1)}{p(x=1)} = H(Y|X)$$

b) $H(Z) \underset{\text{conditionning reduce entropy}}{\geq} H(Z|X) \overset{1.a)}{=} H(Y|X) \overset{X \perp Y}{=} H(Y)$

Similarely $H(Z) \geq H(Z|Y) = H(X|Y) = H(X)$

$(p(z=\alpha, y=\beta) = p(X \oplus Y = \alpha, y = \beta) \ldots)$

c) $H(X) + H(Y) = -\sum_{x \in \mathcal{X}} p_X \log p_X - \sum_{y \in \mathcal{Y}} p_Y \log p_Y = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{X,Y} \log p_X - \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p_{X,Y} \log p_Y$

$= -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{XY} (\log p_X + \log p_Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{XY} \log (p_X \cdot p_Y)$

$H(X,Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p_{XY} \log p_{XY}$; $H(X,Y) = H(X) + H(Y) \Leftarrow p_{XY} = p_X \cdot p_Y$

$\Leftarrow$ X and Y are independant.

c) More Formally we have $H(X,Y) = H(X) + H(Y) - I(X,Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$

$H(X,Y) = H(X) + H(Y) \iff X$ and Y are independant

Problem 2

- $H(p) = -\sum_{x\in\mathcal{X}} p(x)\log_2 p(x) = -\left(\frac{1}{2}\log_2 2^{-1} + \frac{1}{4}\log_2 2^{-2} + \frac{1}{4}\log_2 2^{-2}\right) = \frac{3}{2}$ bits
- $H(q) = -\sum_{x\in\mathcal{X}} q(x)\log_2 q(x) = -\left(3\cdot\frac{1}{3}\log_2\frac{1}{3}\right) \simeq 1.585$ bits
- $D(p\|q) = \sum_{x\in\mathcal{X}} p(x)\log_2\frac{p(x)}{q(x)} = \frac{1}{2}\log_2\left(\frac{\frac{1}{2}}{\frac{1}{3}}\right) + \frac{1}{4}\log_2\left(\frac{\frac{1}{4}}{\frac{1}{3}}\right) + \frac{1}{4}\log_2\left(\frac{\frac{1}{4}}{\frac{1}{3}}\right) \simeq 0.0850$

  $\left(= \log_2|\mathcal{X}| - H(p) \Leftarrow q \text{ uniform}\right)$
- $D(q\|p) = \sum_{x\in\mathcal{X}} q(x)\log_2\frac{q(x)}{p(x)} = \frac{1}{3}\left(\log_2\left(\frac{\frac{1}{3}}{\frac{1}{2}}\right) + \log_2\left(\frac{\frac{1}{3}}{\frac{1}{4}}\right) + \log_2\left(\frac{\frac{1}{3}}{\frac{1}{4}}\right)\right) \simeq 0.0817$

$\left(\frac{1}{3}(1+2+2) - \log_2 3\right)$

We found an example of $D(p\|q) \neq D(q\|p)$

Problem 3

- $D(p\|q) = \sum_{x\in\mathcal{X}} p(x)\log_2\frac{p(x)}{q(x)} = -\sum_{x\in\mathcal{X}} p(x)\log_2\left(\frac{p(x)}{q(x)}\right)^{-1} = -\sum_{x\in\mathcal{X}} p(x)\log_2\left(\frac{q(x)}{p(x)}\right) = -E_p\left[\log_2\frac{q}{p}\right]$

Jensen : $-E_p\left[\log_2\frac{q}{p}\right] \geq -\log_2 E_p\left[\frac{q}{p}\right] = -\log_2\sum_{x\in\mathcal{X}} p(x)\cdot\frac{q(x)}{p(x)} = -\log_2\sum_{x\in\mathcal{X}} q(x) = -\log_2 1$

$(-\log(x))'' = \frac{1}{x^2} > 0 \Rightarrow -\log(x)$ convex

$E_X[g(X)] = g(E_X[X]) \iff g$ affine or X constant $= 0$

because $\sum_{x\in\mathcal{X}} p(x) = \sum_{x\in\mathcal{X}} q(x) = 1$, $D(p\|q) = -E_p\left[\log_2\frac{q}{p}\right] = 0 \iff p(x) = q(x)$ $\left(\frac{q}{p} = 1\right)$

Problem 4 $a_1, \dots a_N \geq 0$, $b_1, \dots, b_N \geq 0$

$$\sum_{i=1}^N a_i\log\frac{a_i}{b_i} = \left(\sum_{j=1}^N b_j\right)\sum_{i=1}^N\left[\frac{1}{\left(\sum_{j=1}^N b_j\right)} a_i\log\frac{a_i}{b_i}\right] = \left(\sum_{j=1}^N b_j\right)\sum_{i=1}^N\left[\frac{b_i}{\left(\sum_{j=1}^N b_j\right)}\left(\frac{a_i}{b_i}\right)\log\left(\frac{a_i}{b_i}\right)\right]$$

$x \quad \log(x)$

$(x\log x)'' = \frac{1}{x} > 0$

$\Rightarrow x\log x$ convex

$$\overset{\text{Jensen}}{\geq} \left(\sum_{j=1}^N b_j\right)\left[\sum_{i=1}^N\frac{b_i}{\left(\sum_{j=1}^N b_j\right)}\left(\frac{a_i}{b_i}\right)\right]\log\left[\sum_{i=1}^N\frac{b_i}{\left(\sum_{j=1}^N b_j\right)}\left(\frac{a_i}{b_i}\right)\right] = \left(\sum_{i=1}^N\frac{\left(\sum_{j=1}^N b_j\right)}{\left(\sum_{j=1}^N b_j\right)}\cdot\frac{b_i}{b_i}\cdot a_i\right)\log\left(\frac{1}{\left(\sum_{j=1}^N b_j\right)}\sum_{i=1}^N\frac{b_i}{b_i}\cdot a_i\right)$$

$$\sum_{i=1}^N a_i\log\frac{a_i}{b_i} \geq \left(\sum_{i=1}^N a_i\right)\log\left(\frac{\sum_{i=1}^N a_i}{\sum_{i=1}^N b_i}\right)$$